Name_____ Date_____ Period_____

LESSON 6-2 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Dividing a Fraction by a Fraction

Maryland Standard 6.NS.A.1 · Learning target – I can divide a fraction by another fraction using the reciprocal.

REMEMBER

- To divide by a fraction, **multiply by its reciprocal**: 3/4 ÷ 3/8 = 3/4 × 8/3.
- Keep the first fraction, change ÷ to ×, and flip only the **second** fraction.
- Check: quotient × divisor = dividend. 2 × 3/8 = 6/8 = 3/4.

WORKED EXAMPLE

How many 3/8-cup servings are in 3/4 cup of yogurt?

Step 1 Write the division: 3/4 ÷ 3/8

Step 2 Multiply by the reciprocal: 3/4 × 8/3 = 24/12

Step 3 Simplify: 24/12 = 2

Answer **2 servings**

WORD BANK

reciprocal – the flipped fraction: 3/8 becomes 8/3

divisor – the number you divide by

quotient – the answer when you divide

PRACTICE

show your work · label units

1. The bar is 1 whole. The shaded part is 4/5 of the bar. How many 1/10 pieces fit in 4/5? Write the division and the quotient.

Step 1 Count the shaded tenths.

Step 2 Each shaded tenth is one 1/10 piece.



1 whole

Division_____

2. Divide. Write each quotient in simplest form.

a. 3/4 ÷ 1/8 = _____ **b.** 2/3 ÷ 2/9 = _____ **c.** 5/6 ÷ 1/3 = _____

d. 7/10 ÷ 2/5 = _____ **e.** 1/2 ÷ 3/4 = _____ **f.** 8/9 ÷ 4/9 = _____

3. **Pancakes.** A recipe needs **2/3 cup** of milk. Rosa has only **1/2 cup**. What **fraction of the recipe** can she make?

   Step 1 Write the division: 1/2 ÷ 2/3.

   Step 2 Flip only the second fraction, then multiply.

   Show your work.

   Answer_____ of the recipe

4. **Bookmarks.** Jae has **5/6 yard** of ribbon. Each bookmark needs **5/12 yard**. How many bookmarks can Jae make?

   Step 1 Write 5/6 ÷ 5/12.

   Step 2 Multiply 5/6 by 12/5, then simplify.

   Answer_____ bookmarks

5. **Find the Error.** Imani wrote 4/5 ÷ 2/3 = 4/5 × 2/3 = 8/15. Explain her mistake. Then find the correct quotient.

   Frame *Imani forgot to ___ the second fraction. The correct answer is ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME With a measuring cup, show someone at home how many 1/4-cup scoops are in 3/4 cup.

Name_____ Date_____ Period_____

LESSON 6-2 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Dividing with Mixed Numbers

Maryland Standard 6.NS.A.1 · Learning target – I can divide mixed numbers by writing them as improper fractions first.

REMEMBER

- Write each mixed number as an **improper fraction**: 2 3/4 = 11/4.
- Then multiply by the **reciprocal** of the divisor.
- Simplify. Write an improper answer as a mixed number.

WORKED EXAMPLE

A board is 5 1/4 feet long. How many 3/4-foot pieces can be cut?

Step 1 5 1/4 = 21/4, so 21/4 ÷ 3/4 = 21/4 × 4/3

Step 2 Multiply and simplify: 84/12 = 7

Answer **7 pieces**

WORD BANK

mixed number – a whole number and a fraction, like 2 3/4

improper fraction – top number is bigger, like 11/4

reciprocal – the flipped fraction: 3/4 becomes 4/3

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. 4 1/2 ÷ 3/4 = _____ **b.** 3 1/3 ÷ 5/6 = _____ **c.** 2 1/4 ÷ 1 1/2 = _____

d. 6 ÷ 1 1/3 = _____ **e.** 5 5/6 ÷ 1 1/6 = _____ **f.** 1 7/8 ÷ 2 1/2 = _____

2. Write each mixed number as an improper fraction.

Step 1 Multiply the whole number by the bottom number.

Step 2 Add the top number. Keep the bottom number.

a. 3 2/5 = _____ **b.** 4 5/6 = _____

3. Soup Bowls. A pot has **8 3/4 cups** of soup. Each bowl holds **1 1/2 cups**. How many bowls can be **filled all the way**? How much soup is **left over**?

Step 1 Change both mixed numbers to improper fractions.

Step 2 Divide. Only whole bowls count.

Step 3 Subtract to find what is left.

Show your work.

Answer_____

4. Hiking. A trail is 7 1/2 miles long. Kenji hikes 1 1/4 miles each hour. How many hours will it take him to hike the whole trail?

Step 1 Write 7 1/2 = 15/2 and 1 1/4 = 5/4.

Step 2 Multiply 15/2 by 4/5, then simplify.

Answer_____ hours

5. Reason. Without dividing, decide: is 6 1/2 ÷ 1 1/4 greater than or less than 6 1/2?

Frame *The answer is ___ than 6 1/2 because 1 1/4 is ___ than 1.*

☐ greater than 6 1/2 ☐ less than 6 1/2

TEST PRACTICE

choose the best answer

6. What is 3 1/2 ÷ 1 3/4?

A 1/2 **B** 2 **C** 3 2/3 **D** 6 1/8

7. A rope is 6 2/3 feet long. It is cut into pieces that are each 5/6 foot long. How many pieces are there?

A 1/8 **B** 5 5/9 **C** 8 **D** 12

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Measure 1 1/2 cups of water with a 1/4-cup scoop. Does the number of scoops match 1 1/2 ÷ 1/4?